BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: 4191

Retyped faithfully from the original board paper — same questions, no changes.

Q. Paper : II (Objective Type)

سوالیہ پرچہ: II (معروضی طرز)

QQ. 1. 1221 — (Inter Part - II) (Session 2017-19 to 2019-21) — Mathematics (Objective) — Paper (II)

Q. Time Allowed:- 30 minutes — PAPER CODE 4191 — Maximum Marks:- 20

Q1. The function $f(x) = (x^3 - x)/(x^2 + 1)$ is _____ function

(A) (B)

(C) (D)

Q2. $\lim \theta \to 0 \ Sin7\theta/\theta =$

(A) (B)

(C) (D)

Q3. $d/dx \ (sec^{-1}x) =$

(A) (B)

(C) (D)

Q4. The derivative of tan x w.r.t. tan x is

(A) (B)

(C) (D)

Q5. $d/dx \ (1/\sin x) =$

(A) (B)

(C) (D)

Q6. $d/dx\ (a^x) =$

(A) (B)

(C) (D)

Q7. $\int \text{cosec}^2 x/\cot x\ dx =$

(A) (B)

(C) (D)

Q8. $\int 1/(9+x^2)\ dx =$

(A) (B)

(C) (D)

Q9. Moment of force $\bar{F}$ about O is

(A) (B)

(C) (D)

Q10. $\int e^x\ (1/x - 1/x^2)\ dx =$

(A) (B)

(C) (D)

Q11. $\int_0^{\pi} \text{Sin}x\ dx =$

(A) (B)

(C) (D)

Q12. The distance of the point (-4, 5) from y-axis is

(A) (B)

(C) (D)

Q13. The slope of the line with inclination 90^0 is

(A) (B)

(C) (D)

Q14. The lines $a_1x + b_1y + c_1=0$ and $a_2x + b_2y + c_2=0$, are perpendicular if

(A) (B)

(C) (D)

Q15. The point (1,0) lies in the solution of the inequality

(A) (B)

(C) (D)

Q16. The centre of the circle $x^2+y^2+2gx+2fy+c=0$, is

(A) (B)

(C) (D)

Q17. The equation of directrix of the parabola $y^2=4ax$, is

(A) (B)

(C) (D)

Q18. Vertices of the hyperbola $x^2/16 - y^2/4 =1$, are

(A) (B)

(C) (D)

Q19. Two vectors $\underline{u}$ and $\underline{v}$ are perpendicular if

(A) (B)

(C) (D)

Q20. $(\underline{i} \times \underline{j}) \times \underline{k} =$

(A) (B)

(C) (D)

Board typos kept verbatim: 'The function f(x) = $(x^3 - x)/(x^2 + 1)$ is _____ function', 'Quadratic', 'Sin7θ', 'Zero', 'The derivative of tan x w.r.t. tan x is', 'Moment of force $\bar{F}$ about O is', 'The distance of the point (-4, 5) from y-axis is', 'The slope of the line with inclination 90° is', 'The lines $a_1x + b_1y + c_1=0$ and $a_2x + b_2y + c_2=0$, are perpendicular if', 'The point (1,0) lies in the solution of the inequality', 'The centre of the circle $x^2+y^2+2gx+2fy+c=0$, is', 'The equation of directrix of the parabola $y^2=4ax$, is', 'Vertices of the hyperbola $x^2/16 - y^2/4 =1$, are', 'Two vectors $\underline{u}$ and $\underline{v}$ are perpendicular if'. CAVEAT: Q1 option (C) right edge cut ('Line...'), Q3 option (C) and Q5 options (A)-(C) obscured/cut in the scan, marked [illegible in source scan].

Paper : II (Essay Type)

پرچہ: II (انشائیہ طرز)

QSection I. 1221SP — Mathematics (Subjective) — (Inter Part - II) (Session 2017-19 to 2019-21) — Paper (II)

Q. Time Allowed: 2.30 hours — Maximum Marks: 80

INCOMPLETE PAPER: the archive scan of the subjective paper contains only page 1 (Section I — Q2 and Q3). The rest (Q4, Section II) is not available; the scan is marked 'P.T.O' and the second page was never archived. This JSON is transcribed faithfully from what exists — nothing invented.

Board typos kept verbatim: 'Determine whether the function f(x)=3x⁴-2x²+7 is even or odd', 'If f(x)=√(x+2) find the domain of f⁻¹(x)', 'Evaluate lim x→∞ (x/(1+x))ˣ', 'If g(x)=(x²+1)², find gog(x)', 'Find dy/dx if y=(2x-3)/(2x+1)', 'If 3x+4y+7=0, find dy/dx', 'Prove that d/dx(cos⁻¹x)= -1/√(1-x²)', 'Find dy/dx if y=cos√x + √sin x', 'If y=e⁻ˣ(x³+2x²+1), find dy/dx', 'Find y₂ if y=x²e⁻ˣ', 'Find extreem values of f(x)=5x²-6x+2', 'If y=ℓn(Tanhx), Find dy/dx', 'Use differentials to approximate (31)^(1/5)', 'Integrate x√(x²-1) w.r.t x', 'Find ∫ dx/(x(ℓn 2x)³)', 'Evaluate ∫ (x+b)dx/(x²+2bx+c)^(1/2)', 'Solve ∫(ℓn x)².dx', 'Evaluate ∫₋₁³ (x³+3x²) dx', 'Solve the differential equation sec²x tan y dx + sec²y tan x dy = 0', 'Evaluate ∫₀^(π/6) x cos x dx', 'Find h so that the points A(4,1), B(2,7) and C(-6,-7) are vertices of a right triangle with right angle at vertix A.', 'Transform the equation 5x - 12y + 39 = 0 into normal form.', 'Find k so that the line joining A(7,3) and B(k,-6) and the line joining C(-4,5) and D(-6,4) are perpendicular.', 'Find the equations of the lines represented by 20x² + 17xy - 24y² = 0.', 'Answer briefly any Eight parts from the followings:-', 'extreem', 'vertix'.